

Assignment 7

Program -

1] Reversing Stack using Queue

```
class Queue
{
    constructor ()
    {
        this.front = null
        this.rear = null
    }

    insert(ele)
    {
        var node = new newNode(ele)
        if(this.front==null)
            this.front=node
        else
            this.rear.next=node
        this.rear=node
    }
    //Get from queue and push on stack
    push_stack()
    {
        var temp=this.front
        while(temp!=null)
        {
            top.push(temp.data)
            temp=temp.next
            this.front=this.front.next
        }
    }
    display()
    {
        var temp=this.front
        while(temp!=null)
        {
            console.log(temp.data)
            temp=temp.next
        }
    }
}
class Stack
{
    constructor ()
    {
        this.top = null
    }

    push(ele)
    {
        var node=new newNode(ele)
        node.next=this.top
        this.top=node
    }
    //pop from stack and add to queue
```

```

    pop_add_queue()
    {
        var temp=this.top
        while(temp!=null)
        {
            front.insert(temp.data)
            temp=temp.next
        }
        this.top=null
    }
    display()
    {
        var temp=this.top
        while(temp!=null)
        {
            console.log(temp.data)
            temp=temp.next
        }
    }
}

class newNode
{
    constructor (data, next)
    {
        this.data = data
        this.next = null
    }
}

const top = new Stack()
const front = new Queue()
top.push(10)
top.push(20)
top.push(30)
top.push(40)
top.push(50)
top.push(60)
console.log("After PUSH...")
top.display()
top.pop_add_queue()
console.log("After adding to Queue...")
front.display()
front.push_stack()
console.log("After REVERSE...")
top.display()

```

Output -

After PUSH...

60

50

40

30

20

10

After adding to Queue...

60

50
40
30
20
10
After REVERSE...
10
20
30
40
50
60